

JMMC Data science

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JMMC @ OSUG

JMMC: liens

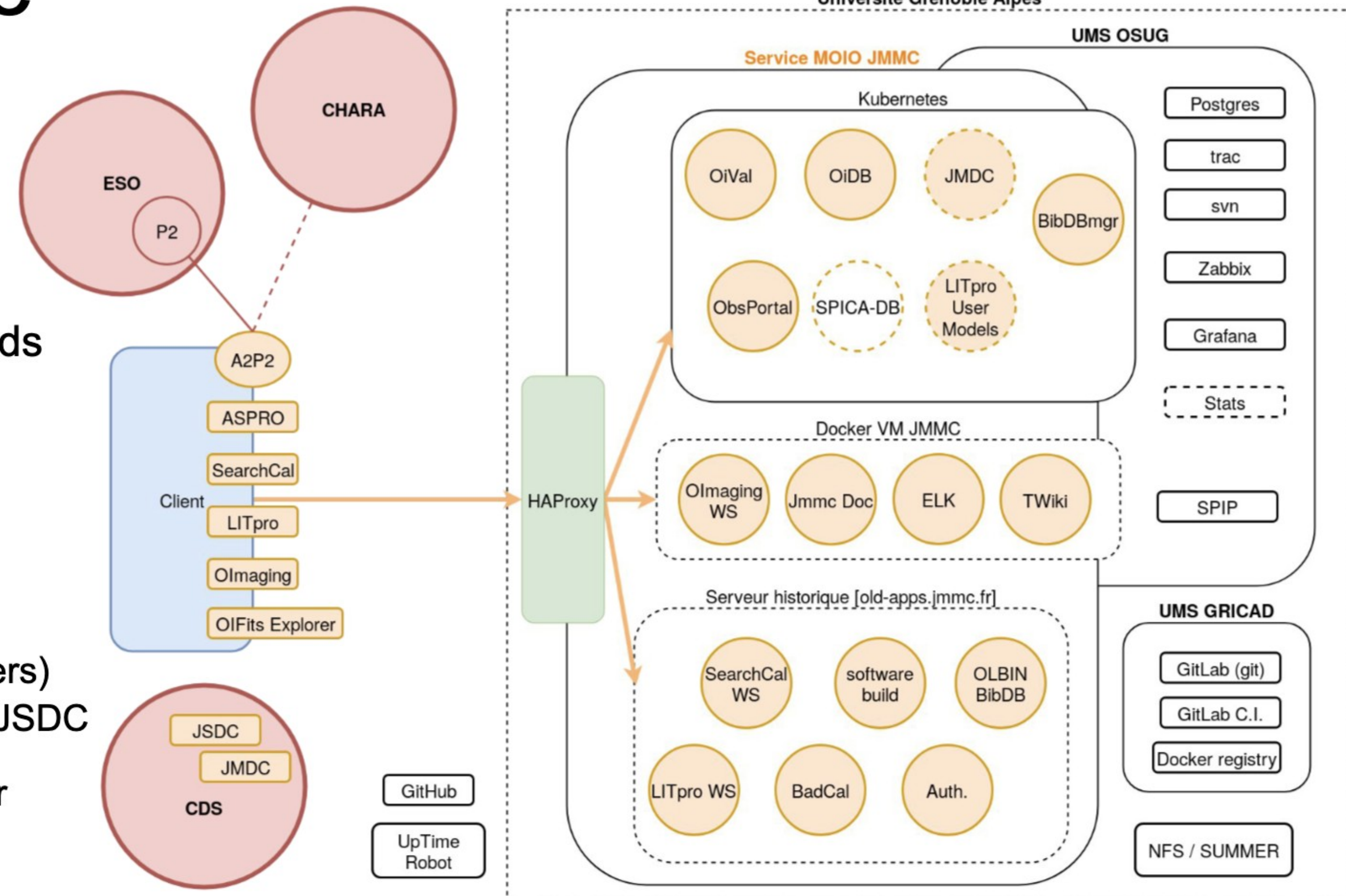
- **web:** <https://www.jmmc.fr/>
- **Releases:** <https://releases.jmmc.fr/index.html>
- **OpenDev:** <https://github.com/JMMC-OpenDev/>

Services:

- **Catalog API:** <http://oidb.jmmc.fr/exist/apps/catalogs/index.html>

Infrastructure @OSUG

- Datastore : 2 TB
- HAProxy : ~ 50 backends
- 6 VMs
 - 8 containers
- 15 pods K8S
- Databases :
 - 5 postgres databases (OSUG) + 1 mysql (users)
 - New SPICA, JMDC & JSDC catalogs stored on the shared postgres server



Encore beaucoup de services à maintenir et transférer dans la nouvelle infrastructure

Aspro2 : preparing proposals



Object Modelling

Calibration

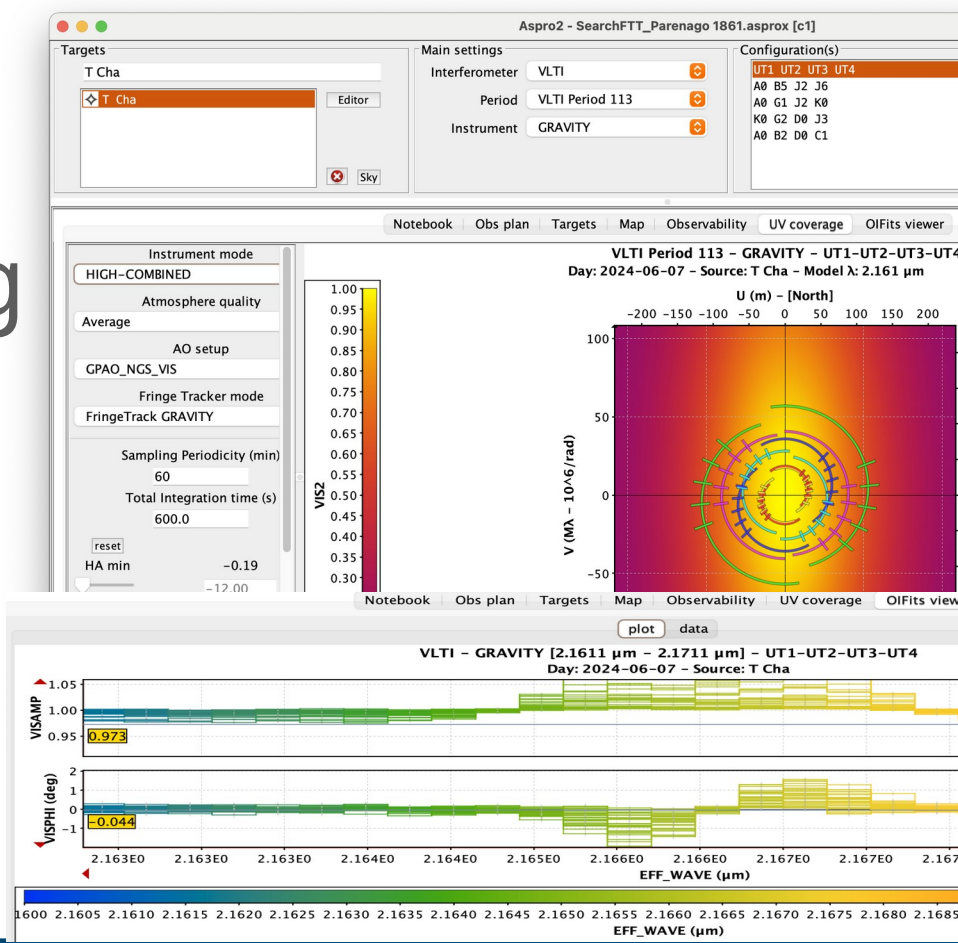
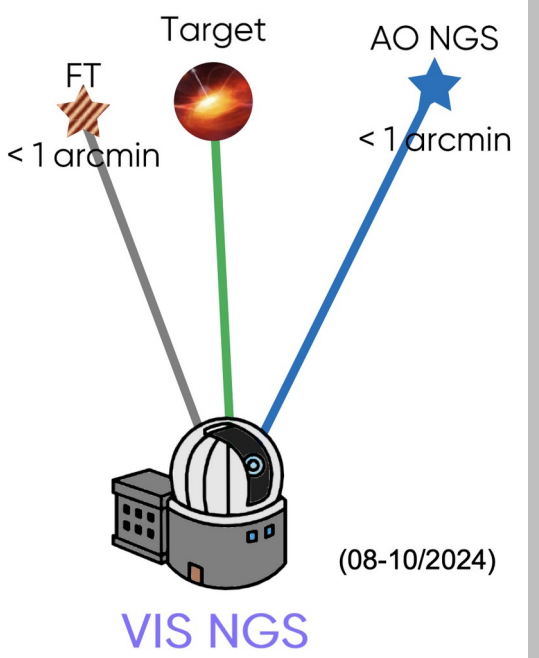
a2p2

Obs Portal

- Analytical
- User provided
- AMHRA Db

Checking previous observations

Simulating observability



Phase 2 v2.20.28

Details Overview Schedule Execution Sequence Help

UT: 09:41:50 · LST: 22:05:

Your Observing Runs

Sort by: Nothing selected

+ Test a2p2 0

+ HD_169142 9

+ TESTGM 8

+ HD_169142 1

+ Parenago_1861 1

+ OB 3945305 · SCI_Parenago_1861_GRAVITY_LOW-COMBINED

OB CB Fld G T C

60.A-9108(A) · GRAVITY · OB 3945305 SCI_Parenago_1861_GRAVITY_LOW-COMBINED

(P)artially Defined

Obs. Description Target Constraint Set Time Intervals

GRAVITY_dual_onaxis_acq

#1 acquisition 2665712

FringeTracker mode

Mode for Metrology Laser

SC object picking strategy

FT object name

FT object K band magnitude

FT object H band magnitude

FT object diameter (mas)

FT object expected visibility

name

K band magnitude

diameter (mas)

AUTO

ON

F

V* V348 Ori

7.989

8.45

0

1

Parenago 1861

9.83

0

Get Star

Search FTT

Welcome to the JMMC GetS!

Please enter your object name

Name: HD 100546

Advanced parameters: show / h

Reset Submit

It may take few seconds to ge

Send VOTable (samp)

GetStar result:

GetStar software (In case of problem, please report to jmmc-user-supp
Request parameters: -objectName HD 100546 -format vot
Generated on (UTC): 2024-11-17T12:00:16

Property
HD
HIP
DM

JMMC SearchFTT

This tool searches for nearby stars suitable for off-axis Fringe Tracking and off-axis Adaptive Optics. You can query one or several Science Targets separated by semicolon by names (resolved using Simbad for proper motion) or by coordinates (RA +J-DEC J2000). For each of them, suitable solution candidates for the FT (infrared) and for the AO (visible). Only solutions with a valid AO and a valid FT are presented in the main output table. For debug or further investigations, the list of all individu extended raw result.

For the time being, only the NGS mode of GPAO is supported by the scoring. However, solutions can be found for the LGS mode by increasing the AO magnitude (but with wrong scoring and ranking SIMBAD and Gaia DR3 catalogues are cross-matched though CDS and ESA data centers. Each query is performed within a search radius around the Science Target below a max declination. A magn: candidates for the FT (infrared) and for the AO (visible). Only solutions with a valid AO and a valid FT are presented in the main output table. For debug or further investigations, the list of all individu extended raw result.

Parenago 1861

Catalogs to query: Gaia DR3 Simbad

Constraints: FT mag: 12 AO mag: 12.5 declination: 40

Submit my identifiers Reset

1 targets with both FT and AO solutions votable

Column visibility CSV Copy

user_identifier	ra	dec	FT identifier	AO identifier	Score	Rank	FT_mag	scf_ft_dist	ao_mag	scf_ao_dist	ft_ao_dist	Catalog
Parenago_1861	83.815	-6.380	V* V348 Ori	V* V348 Ori	0.263	1	7.989	8.162	12.260	8.162	0	Gaia DR3
Parenago_1861	83.815	-6.380	Parenago_1861	V* V348 Ori	0.242	2	8.828	0.000	12.260	8.162	8.162	Gaia DR3
Parenago_1861	83.815	-6.380	COUP_742	V* V348 Ori	0.141	3	10.156	5.912	12.260	8.162	10.979	Gaia DR3
Parenago_1861	83.815	-6.380	*tet01 Ori B2	V* V348 Ori	0.090	4	6.004	20.599	12.260	8.162	12.736	Gaia DR3

Query Parameters

1) Instrumental Configuration Magnitude Band: K Wavelength (μm): 2.2 Max. Baseline (m): 102.45

2) Science Object Name: HD 100546 RA 2000 [hh:mm:ss]: 11 33 25.4408872296 DEC 2000 [+/-dd:mm:ss]: -70 11 41.241297948 Magnitude [K]: 5.418

3) SearchCal Parameters Min. Magnitude [K]: 3.418 Max. Magnitude [K]: 7.418 Scenario: Bright Faint RA Range [mm]: 240.0 DEC Range [deg]: 20.0

Get Calibrators

Progress: Found Calibrators (4957 sources, 4454 filtered)

Index	dist	HD	RAJ2000	DEJ2000	vis2	vis2Err	diam_chi2	LDO	e_LDO_ref	UD_V	UD_J	UD_H	UD_K	GroupSize	SIMBAD	SpType
1	0.0	100546	11 33 25.4408872296	-70 11 41.241297948	0.090	0.000	149.211	0.000	0.000	0.000	0.000	0.000	0.000	0	HD 100546	A0VneK...
2	0.791	101331	11 40 20.4632073387	-69 40 18.25368796	0.932	0.011	Relaxed 0th-order of the LDO diameter estimation	0.74	0.745	0	0	0	0	0	HD 101331	K1III
3	0.882	101802	11 42 55.0739668438	-69 50 55.474924888	0.978	0.001	0.802	0.434	2.448	0.394	0.414	0.414	0.418	0	HD 101802	K3III
4	1.165	102412	11 46 45.2447441962	-70 30 49.97939992	0.898	0.014	1.843	0.965	6.913	0.868	0.915	0.915	0.923	0	HD 102412	K5III
5	1.722	27624	11 13 09.1447491760	-70 06 48.421441187	0.983	0.377	2.41	0.344	0.362	0.362	0.364	0	0	0	HD 27624	K1III
6	1.765	27202	11 14 25.0693954131	-69 51 52.382857655	0.983	0.002	1.518	0.566	2.446	0.513	0.541	0.541	0.545	0	HD 27202	K3III
7	1.771	101386	11 53 59.0351546027	-69 56 09.213846011	0.985	0.001	2.777E-4	1.717	0.357	2.376	0.327	0.343	0.343	0	HD 101386	K0III
8	2.259	28811	11 21 15.6185658855	-68 12 34.883210689	0.919	0.015	0.654	0.865	8.738	0.768	0.811	0.811	0.819	0	HD 28811	M0III
9	2.283	28581	11 19 27.4013671228	-68 16 30.761501493	0.977	0.001	0.384	0.443	2.243	0.405	0.425	0.425	0.428	0	HD 28581	K1III
10	2.353	28513	11 18 55.6486964549	-68 13 30.095101851	0.985	0.001	2.2E-4	0.868	0.355	2.376	0.324	0.341	0.343	0	HD 28513	K1III

Le cycle de la donnée

Optical interferometry DataBase



Target name or position

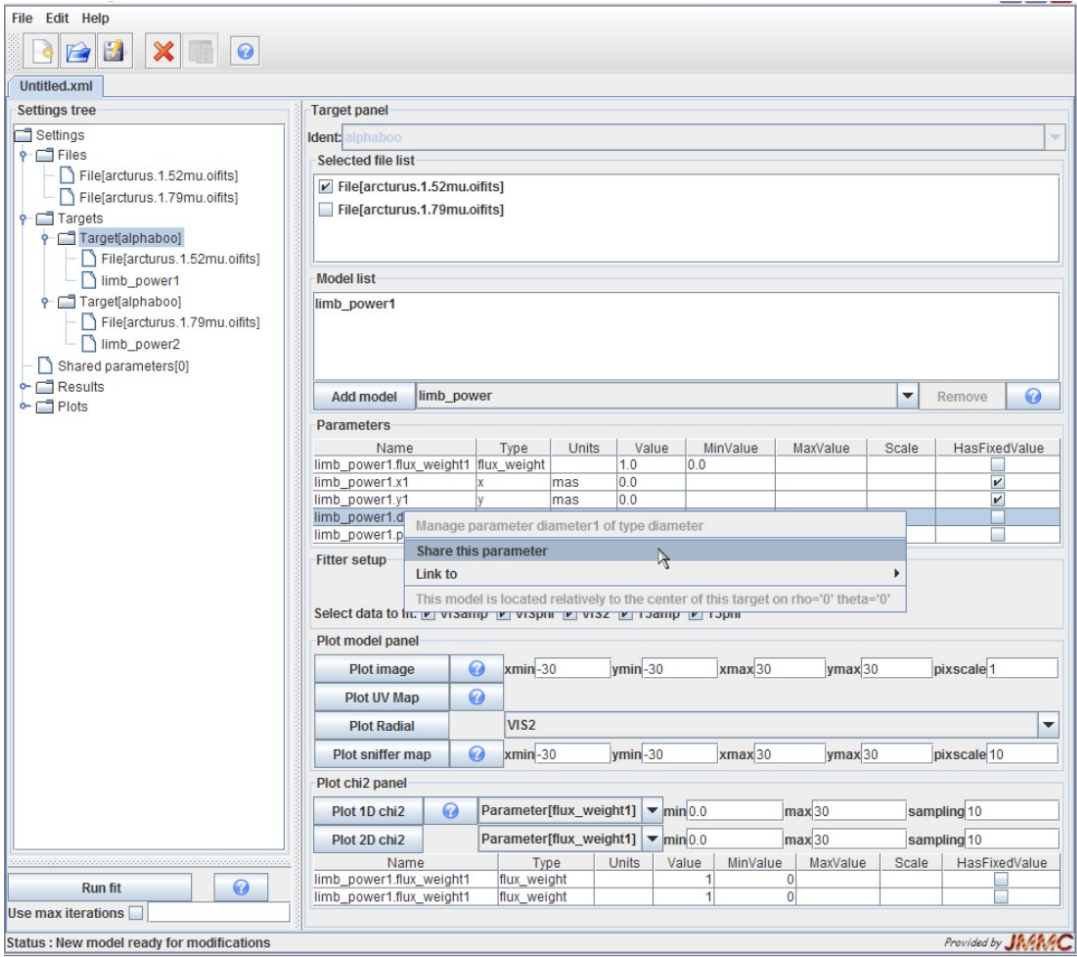
Q

Enter target name or visit the advanced form

Please use next url to use OiDB's TAP server : <http://tap.jmmc.fr/voltt/tap/>

Data Validation Exploration

LitPro



JMMC

OImaging

Why share your data on OiDB?

Preserve your reduced data

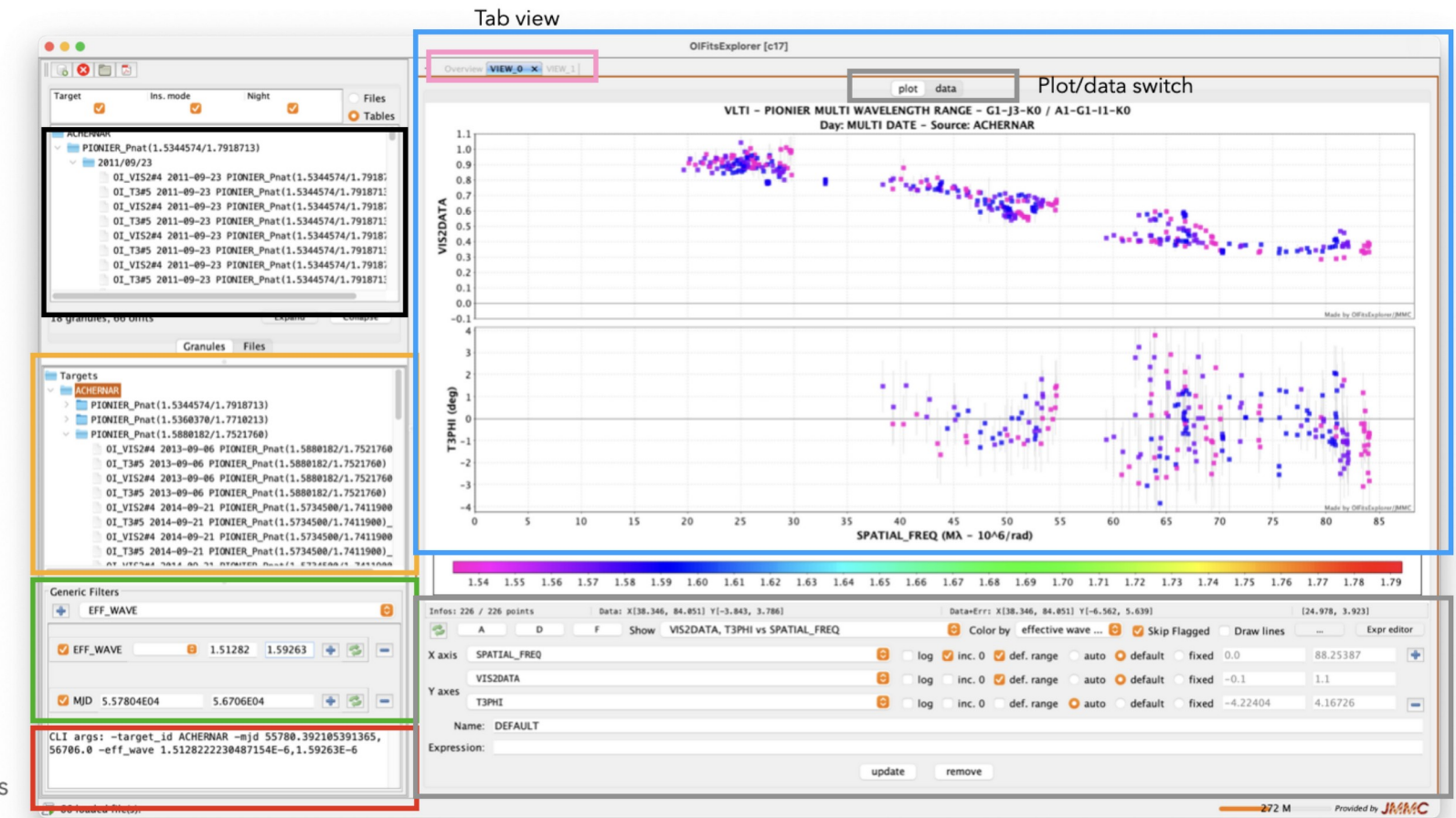
Increase your citation rate

Checking previous observations

Capitalize on the data reduction

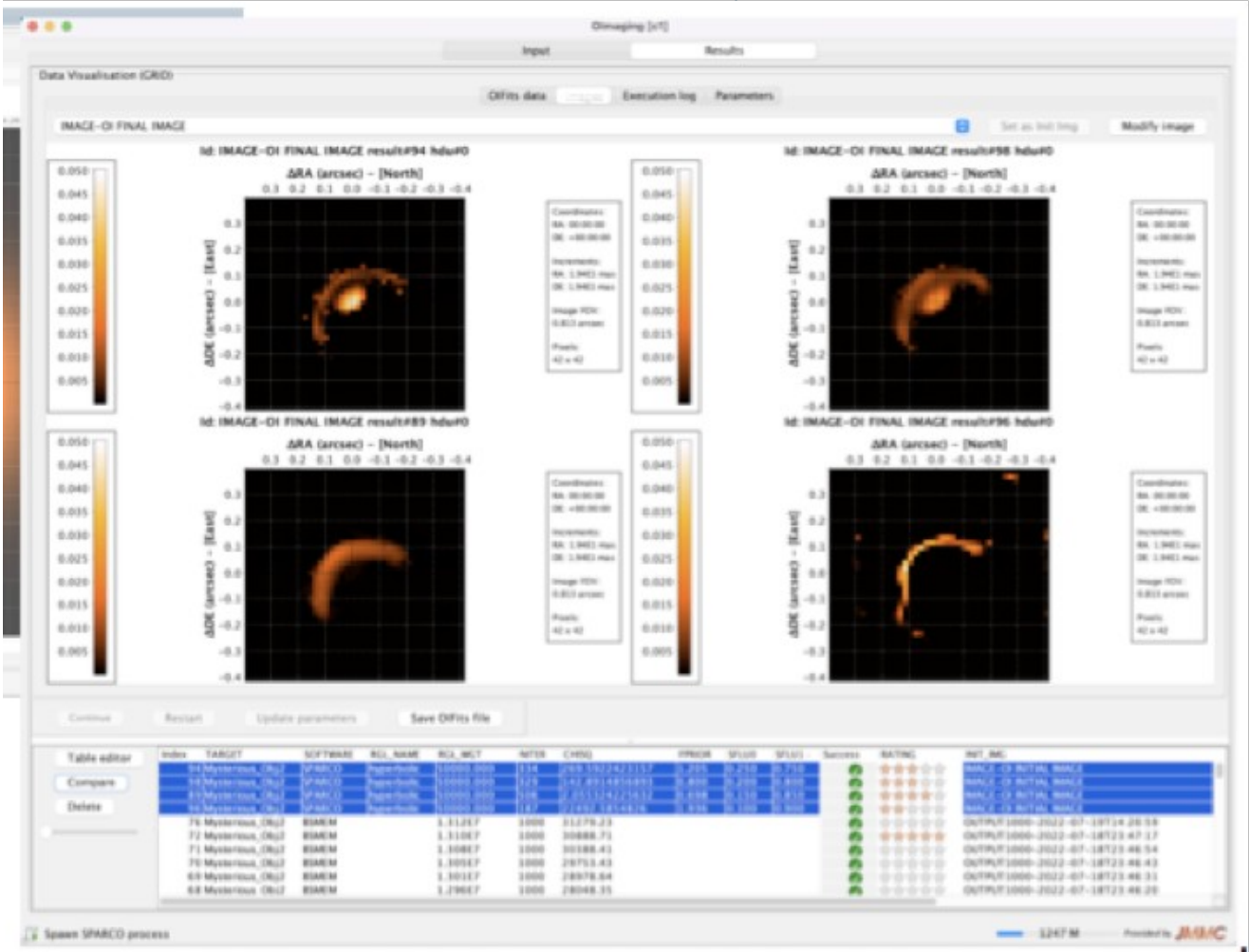
Interoperability VO protocols and tools

Create collaborations



Plotting parameters

Model fitting



OifitsExplorer

Conclusion

- More catalogs in tap.jmmc.fr
 - JMDC
 - JSDC3 ...
- Improve python wrapper / notebook pour simplifier l'accès aux services et aux données ...